

Towards Semantic Scene Segmentation for Autonomous Agricultural Vehicles

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Abstract—The development of autonomous agricultural vehicles depends on accurate visual interpretation of the environment. Semantic pixel-wise segmentation plays an important role in achieving a more detailed description of the scene and understanding of the spatial-relationship between different objects. In this paper, a deep architecture for road scene and indoor scene segmentation, SegNet, is applied in solving the scene recognition problem for the agricultural environment. The network is trained on an agriculture image dataset collected during field operation. The problem focuses on the segmentation of five classes of structures commonly found in the fields: vegetation, grass, ground, crop field and obstacles. We apply frequency balancing to address the challenge of class imbalance and transfer learning technique. The quantitative performance of the network is evaluated using pixel accuracy and intersection over union metrics. Moreover, we present an approach for an infield qualitative validation of the trained network.

I. INTRODUCTION

According to the latest UN World Population Prospects, the world's population is projected to reach 9.7 billion by 2050 and 10.9 billion by 2100 [1]. In order to meet the increasing food demands and provide sustainable agricultural growth, it is essential to implement the more efficient use of land, labour and other inputs through technological progress, social innovation and new business models [2].

The future vision of sustainable agricultural growth relies on technological innovation in the areas of precision agriculture, robotic systems and information technologies [3]. The recent availability of large-scale datasets, use of GPU computing, and increased computing power led to the application of AI technologies in a wide range of businesses and industries.

Current applications of AI in agriculture are focused on precision farming and include crop yield estimation, monitoring of crop health, weed detection, and predictive weather monitoring.

The demand for AI in agriculture is also present in the development of autonomous agricultural vehicles. Today, a number of self-driving agricultural vehicles is in the development [4],[5],[6],[7]. However, their current capacity to understand the surrounding environment is insufficient for fully autonomous and safe execution of complex field tasks. During driving in the field, an agricultural vehicle

must also continuously perform a task (harvesting, seeding, spraying, bailing, etc.) in a diverse and highly unstructured environment. Therefore, the development of accurate and robust perception system is vital in order to provide safe and reliable operation for autonomous agricultural vehicles.

The vehicle's ability to perceive and understand its environment relies heavily on the processing of visual inputs. The recent success of deep learning architectures for image classification, semantic segmentation, and object detection inspired application of deep learning in scene perception for autonomous vehicles. Semantic pixel-wise segmentation is an important part of the scene perception task. The goal of semantic segmentation is to annotate each pixel within an image with a semantically interpretable class from a predefined set of classes. This annotation helps achieve a detailed description of the object boundaries and better understanding of the spatial relationship between different structures present in the scene.

This research work aims to advance the real-life application of semantic segmentation in scene perception for the agricultural environment. The SegNet encoder-decoder architecture [8] is applied in solving the segmentation task. This architecture was selected due to its practical advantages. Compared to architectures presented in [9] and [10], SegNet provides a memory-efficient storing of information from encoder feature maps by using only max-pooling indices. The work presented in [11] uses a similar technique for upsampling in the decoder network, but it has a significantly larger number of parameters. Moreover, SegNet is a fully convolutional network and can therefore take input images of arbitrary size. To the best of our knowledge, SegNet architecture has not been applied before in solving scene segmentation task for agricultural vehicles.

The main barrier in transferring semantic pixel-wise segmentation architectures to agriculture is limited availability of labelled datasets [12]. Therefore, an important contribution of our work is training and performance evaluation of the segmentation network on a new image dataset collected specifically for the agricultural domain. The images were collected during tractor's operation in the crop field and represent a realistic scenario for driving in the agricultural environment. The dataset contains five labelled classes of field structures: "Vegetation", "Grass", "Ground", "Crop field", and "Obstacle".

Moreover, we present an approach for continuous qualitative validation of the network in practice. The approach is based on a tool for real-time visualization of segmentation results for images captured by the vehicle's camera during

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field operation. This approach will enable us to recognize the challenging scenarios in the field and collect data that is relevant for further improvements of the network more efficiently.

The remainder of the paper is structured as follows. In II related work in deep learning based segmentation is reviewed. Also, the application of deep learning and AI in the agricultural domain, for solving various tasks, is reviewed. The SegNet architecture and the training process are described in more detail in III. In IV, the performance of the trained network is evaluated. This is followed by the conclusion in V.

II. RELATED WORK

In recent years, semantic pixel-wise segmentation gained significant attention from machine learning researchers. Accurate and efficient image segmentation is needed in many core computer vision problems, such as autonomous driving, medical sciences, human-machine interaction, image search engines and augmented reality [13].

Many segmentation architectures are inspired by the contributions and success of standard architectures like AlexNet [14], GoogLeNet [15], VGG [16], and ResNet [17]. The Fully Convolutional Network (FCN) presented in [9] is considered a breakthrough in deep learning based semantic segmentation. Unlike previously mentioned architectures, FCNs consists of convolutional layers without any fully-connected layers. Pixel-wise prediction and recovery of spatial information are achieved by upsampling using deconvolutions. The important feature of FCNs is their ability to make pixel-wise predictions for inputs of arbitrary size [9].

SegNet is an efficient encoder-decoder neural network for pixel-wise segmentation. The encoder network in SegNet is topologically identical to the convolutional layers in VGG16 and without fully connected layers. The key component of SegNet is the decoder network that uses a set of decoders each corresponding to one encoder. Unlike architectures that store the full encoder network feature maps, SegNet only stores the max-pooling indices of the feature maps and uses them in its decoder network for upsampling. The storing of indices provides good performance and efficiency in terms of memory and computational time. Compared with other relevant architectures, SegNets perform competitively on well-known datasets and are one of the best models for solving scene segmentation problems [8].

The U-Net [10], initially proposed for biomedical image segmentation, is a fully convolutional encoder-decoder network with symmetric architecture. Unlike SegNet, U-Net stores entire feature maps. The upsampled decoder features are concatenated with corresponding high resolution encoder feature maps for better localization. DeconvNet [11] encoder consists of convolutional layers and two fully connected layers of VGG16. Compared to SegNet, DeconvNet have a larger number of parameters and longer training and inference time.

In agriculture, image segmentation has been applied in solving a wide range of tasks related to precision farming,

automation and autonomy.

One of the applications is the detection of field boundaries. In [18], a deep convolutional neural network for semantic segmentation, ResUNet, is used for field boundary extraction from satellite images. The study presented in [19] used DeepLabv3+ architecture to classify and extract crop areas from high spatial resolution remote sensing images.

Deep learning based crop and weed segmentation gained attention in recent years. The architecture presented in [20] relies on an FCN with an encoder-decoder structure and incorporates spatial information by considering image sequences to compute the label mask. The architecture inspired by SegNets for segmentation of crops, weeds, and soil, implemented on an agricultural robot, is presented in [21]. In [22], a modified version of SegNet is used to segment crop and weed from multispectral images captured by an unmanned aerial vehicle.

Semantic segmentation for obstacle detection in agriculture, presented in [23] uses FCN model pre-trained on Pascal-Context dataset. The available general classes are mapped to classes of a new agriculture specific dataset. The agriculture dataset is collected in dairy field and various orchards fields in Australia. The work presented in [24] introduces DeepAnomaly, an algorithm that combines deep learning and anomaly detection for the detection of occluded and unknown objects in the agricultural environment. This approach does not provide labels for detected obstacle but can rather be used as a region proposal algorithm for a classification network. The study presented in [12] uses FCN trained on datasets that do not contain agricultural specific images and classes, ImageNet and Pascal-Context datasets. The classes in the dataset are remapped to a subset of agriculture specific class. The network is tested on a set of 10 images for row crop operation and grass mowing use cases. [25] presents Adversarial Domain Adaptation method for semantic segmentation of agricultural scene. The segmentation network is based on U-Net with the modified residual blocks.

III. ARCHITECTURE

SegNet architecture, illustrated in Fig. 1, consists of encoder network, corresponding decoder network, and softmax classifier. The proposed topology for encoder network corresponds to the first 13 convolutional layers of VGG16 network. Therefore, it is possible to initialize the encoder network with pre-trained weights for large datasets. The encoder network is followed by the decoder network with the same number of convolutional layers. The final soft-max layer produces class probabilities for each pixel independently.

Each encoder layer in the network consists of convolutions, batch normalization and ReLU non-linearity, followed by non-overlapping max-pooling layer (2×2 window and stride 2). The application of the max-pooling layers results in the loss of spatial resolution for the deepest feature maps. Therefore, the max-pooling indices of each low-resolution feature map in the encoder network are stored and used in the upsampling of the corresponding input feature map in

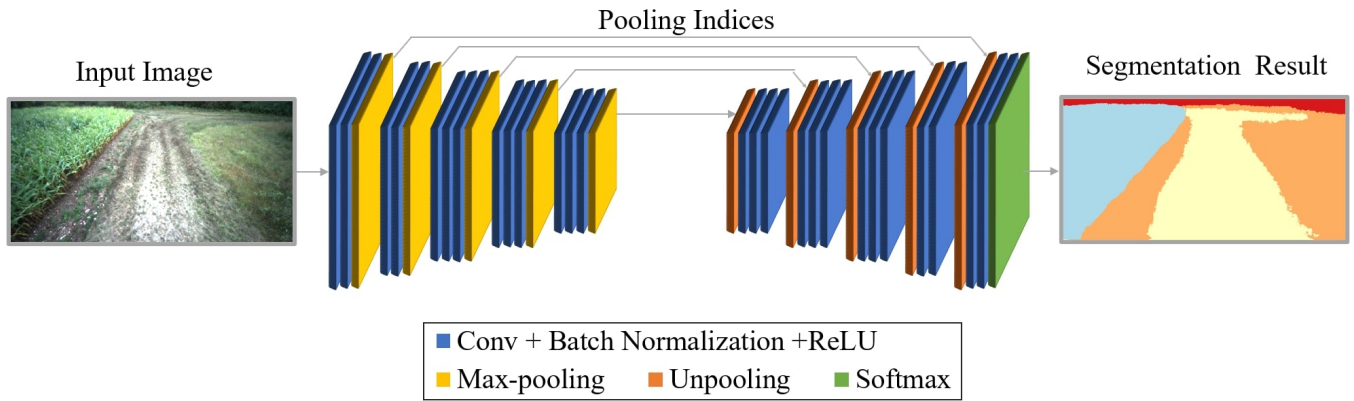


Fig. 1: Illustration of SegNet architecture applied to a scene segmentation task in agricultural environment. The input of the network is an RGB image of a field scene. The output is the corresponding pixel-wise segmented image.

decoder network. The upsampled feature maps are convolved with trainable filters to produce dense feature maps. The input to the network is 3-channel RGB images where each channel corresponds to one of the components red (R), green (G) or blue (B). The decoder network produces a multi-channel feature map. This feature map is fed to a soft-max classifier that produces a pixel-wise classification. The output of the soft-max classifier, for N classes, is an N channel image of class probabilities. The predicted class for a specific pixel is the one with the maximum probability.

The SegNet architecture used in this paper consists of 5 encoder-decoder pairs followed by a soft-max classifier. The convolutions in the five encoder layers have 64, 128, 256, 512 and 512 filters respectively, with 3×3 kernel size, the stride length of 1 and padding equal to 1. The convolutions in the decoder layers coincide with the ones from the encoder pair. The output of the soft-max classifier is a 5-channel image of pixel-wise probabilities for the five classes stated in I.

A. Training

The network was trained on an agriculture dataset collected in Denmark during two data collection campaigns in summer and winter season. The dataset consists of 765 RGB images, with corresponding ground truth images, at 516×258 pixels resolution. The five labelled classes are described in Table I.

The dataset has been randomly split into training, validation and test set. Approximately 15% of images are used for testing, and the remaining images are split into testing and validation set with 75% and 25% ratio. The data loaders for training and validation datasets are collating data samples into batches of 5 images. At the beginning of each epoch, the data loaders are reshuffling the order of samples to avoid learning from the input sequence pattern and improve the accuracy on testing set and convergence rate.

The pixel-wise cross-entropy loss is used as the cost function for training the network. The loss is evaluated for each pixel individually and averaged across observations for each batch. In the real-life field scenes, the variation in the number of pixels that belong to each class is significant.

TABLE I: Description of classes in the dataset.

Class	Description
Vegetation	shelterbelts, trees, bushes
Grass	areas covered with grass and other small plants
Ground	areas with standing crop (corn, wheat, etc.)
Crop field	areas with standing crop (corn, wheat, etc.)
Obstacle	fence, pole, rock, building, etc.

This variation is present in our dataset as well, as shown in Fig. 2. Therefore, it is necessary to weight the loss differently for individual classes. We used median frequency balancing as presented in [26]. The weight assigned to each class is equal to the median of all class frequencies divided by class frequency. This balancing results in the larger classes having weights smaller than 1 and the smaller classes are having weights larger than 1. The unlabelled pixels are ignored during training.

Since the convolutions in the encoder network are identical to VGG16, the encoder weights are initialized using VGG16 pre-trained weights on the ImageNet classification challenge dataset. The decoder weights are initialized according to [27].

The encoder-decoder network is trained using the modular training technique presented in [28]. First, the outermost encoder-decoder pair is trained. Then, deeper pairs are added and trained one by one while the weights of the preceding pairs remain fixed. The training is using stochastic gradient descent optimization with learning rate equal to 10^{-2} . During the training of each pair, learning rate scheduling is applied. The implemented scheduler reads training loss values after each epoch, and if no improvement is seen for 10 epochs, the learning rate is reduced to 10^{-3} . Next time the training loss shows no improvement for 10 epochs the training of that pair is stopped, and the next encoder-decoder pair is added and trained. This training approach requires approximately 680 epochs for the training of the whole network. In order to keep the model weights smaller and reduce overfitting, the L2 penalty is added with weight decay parameter of 10^{-4} .

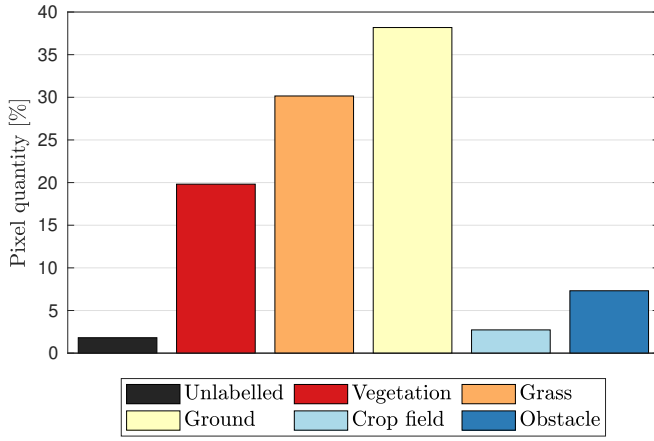


Fig. 2: Illustration of class imbalance in the dataset. The graph shows the percentage of pixels belonging to each class of the dataset.

IV. RESULTS

The performance of the network is evaluated on 100 labelled images using pixel accuracy and IoU metrics. The evaluation is performed by computing a confusion matrix and metrics for each class and across all images in test set. The unlabelled pixels were excluded from the analysis.

Pixel accuracy for each class is calculated as a number of correctly classified pixels of that class, divided by the total number of pixels of that class in the ground truth. That is,

$$pixel_accuracy = \frac{TP}{TP + FN} \quad (1)$$

where TP is the number of true positives and FN is the number of false negatives.

The pixel accuracy of each class, calculated as the average of class accuracy in all images, is listed in Table II. The average class accuracy on the dataset is 73.4%.

In the case of extreme class imbalance, such as ours, pixel accuracy is biased and might provide misleading results. Therefore, IoU metric is considered as well.

The IoU metric is one of the most commonly used metrics in semantic segmentation. For each class, this metric measures the number of pixels common between the ground truth and network segmentation divided by the total number of pixels present in both. That is,

$$IoU = \frac{TP}{TP + FP + FN} \quad (2)$$

where TP is the number of true positives, FP is the number of false positives and FN is the number of false negatives.

The IoU metric is calculated for each class separately and then averaged over all classes to provide the mean IoU. The accuracy for individual classes is listed in Table II. The mean IoU on the test set across all classes is 67.8%.

The results suggest that pixel accuracy is affected by class imbalance and gives slightly better results compared to IoU. For the individual classes, both metrics show good accuracy results for “Vegetation”, “Grass”, and “Ground”

TABLE II: Network performance quantified for each class using pixel accuracy and IoU.

Class	Pixel accuracy	IoU
Vegetation	87.52%	80.74%
Grass	76.75%	71.94%
Ground	81.30%	76.32%
Crop field	54.83%	47.06%
Obstacle	66.53%	62.77%

class, whereas “Crop field” and “Obstacle” classes have somewhat lower accuracy. Referring to data shown in Fig.2, it can be seen that “Crop field” and “Obstacle” classes have significantly lower pixel quantity compared to the other three classes. This indicates that the extreme class imbalance led to lower performance results for the smallest classes, despite the applied class balancing.

Examples of the segmentation results produced by the network on the test images can be seen in Fig. 3. The figure shows the input RGB images together with corresponding ground truth, network segmentation, and the input image overlapped with network segmentation. The rows 1, 3 and 4 show network’s performance in classifying the examples of “Obstacle” class. In the 1st row, we can see a pole and a rock protruding from the grass being classified as “Obstacle”. In the 4th row, the obstacle is a wall of a shed, and the adjacent bush is clearly distinguished from surrounding grass as “Vegetation”. Rows 2 and 5 show segmentation of “Crop field” class. In both examples, the crop field is field of standing corn. The driving paths are classified as “Ground”, and shelterbelt at the edge of the field is classified as “Vegetation”.

The presented results are supplemented with the design of a tool that enables infield qualitative performance evaluation of the trained network. The concept of the tool is illustrated in Fig. 4. The main components of the tool are camera module, SegNet network, and HMI screen. The RGB images are fed to SegNet network at 5Hz and displayed together with the segmented image on the screen. This visualization enables vehicle operator to monitor network performance while driving in the field.

The tool has been successfully tested in practice on an agricultural vehicle equipped with an RGB camera, computer with GPU, and a screen.

The benefit of this approach is the ability to evaluate the model’s performance in real-time and easily identify situations in which the model is failing. These benefits will give us more insight into the current deficiencies of our model and lead to continuous improvements.

V. CONCLUSION

This paper presented the application of SegNet architecture in solving scene segmentation problem for the agricultural environment. The network was successfully trained on our agriculture dataset and showed satisfying results on the test set. In evaluation with two different metrics, the trained network showed good performance for all classes. However,

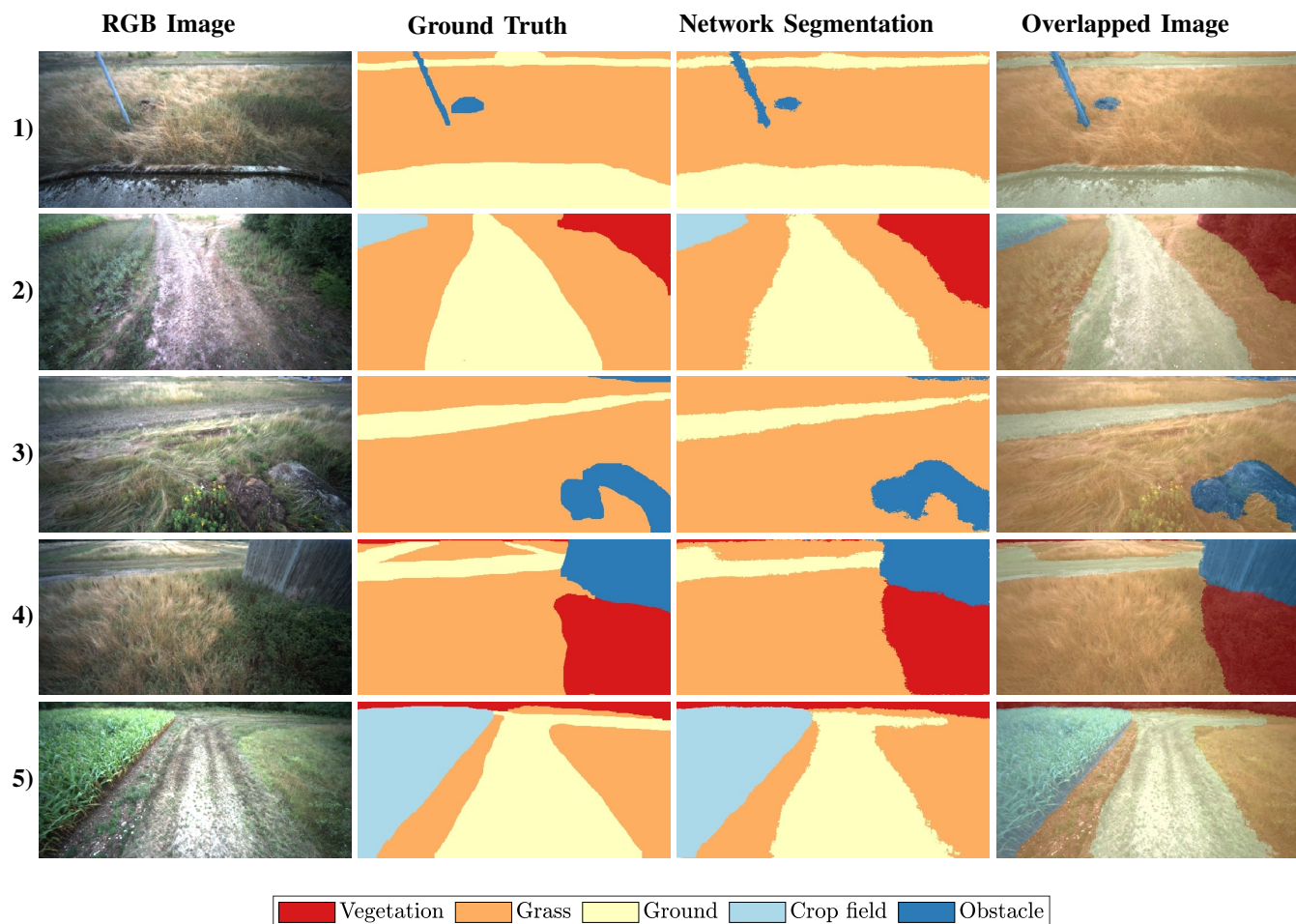


Fig. 3: SegNet results on test set images. Each row corresponds to one example and shows RGB image used as the network input, ground truth image, network segmentation, and input image overlapped with network segmentation.

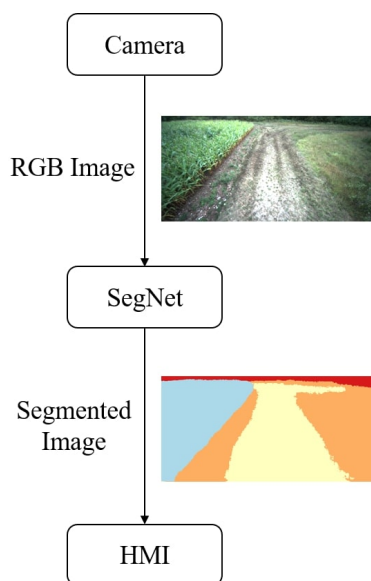


Fig. 4: Concept of the tool for qualitative validation.

the performance results were lower for the two smallest classes.

The presented tool for infield qualitative performance evaluation of the trained network was implemented and tested in the field. The preliminary results were promising. We believe this tool will help us identify the challenging scenarios for our network and give insight into data required for improving the network's robustness and accuracy in the future.

Future work will investigate different strategies for class frequency balancing. The used dataset is relatively small, and training of robust segmentation architecture will require collecting more data and adding variety to the dataset. Therefore, future steps will include collecting more images, adding more classes specific to the agricultural domain, and adding variety to the collected data in terms of locations, seasons, field operations, and weather conditions.

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